Name : …… Student ID: …….

The University of New South Wales

School of Electrical Engineering and Telecommunications

Mid-term Examination (Afternoon)

Term 3, 2021

ELEC3104: Digital Signal Processing

Candidates should attempt the questions as per the following Instructions

1. This mid-term exam constitutes 20% of the total course assessments
2. **Time allowed:** 2 hours (+ 10 min reading time + 20 minutes uploading answers)
3. You must answer a maximum of 5 questions (including the MATLAB Question)
4. **Except** the MATLAB question, all other questions are presented in the SOLO Modified Taxonomy form:
   - The exam will be in the following layout:
     - ❖ Four questions under SOLO - Level 2
     - ❖ One question under SOLO - Level 3
     - ❖ One question under SOLO - Level 4
     - ❖ One question under SOLO - Level 5
   - In doing the exam, you can choose which questions you want to complete, depending on what Level you are aiming for:
     - ❖ If you are aiming to achieve **Level 2**, you will need to **answer all four questions under Level 2** + the MATLAB question. The maximum mark you can achieve is **64.**
     - ❖ If you are aiming to achieve **Level 3**, you will need to answer **any three Level 2 questions and one Level 3 question** + the MATLAB question. The maximum mark you can achieve is **74**.
     - ❖ If you are aiming to achieve **Level 4**, you will need to answer **any two Level 2 questions, one Level 3 question and one Level 4 question** + the MATLAB question. the maximum mark you can achieve is **84**.
     - ❖ If you are aiming to achieve **Level 5**, you will need to **answer any one of the Level 2 questions, one Level 3 question, one Level 4 question and one Level 5 question +** the MATLAB question. The maximum mark you can achieve is **100**.
5. Print your name, student ID and question number on the front page of each **answer** page.
6. The questions are NOT of equal value. Part marks are as indicated.
7. Authorised examination materials: This is an open book exam.
8. This paper contains 11 pages apart from this instruction page.

ANSWERS MUST BE WRITTEN IN INK. EXCEPT WHERE THEY ARE EXPRESSLY REQUIRED, PENCILS MAY ONLY BE USED FOR DRAWING, SKETCHING OR GRAPHICAL WORK. UPLOAD AN IMAGE OF EACH ANSWER INDIVIDUALLY ON MOODLE WITHIN THE ALLOCATED TIME

SOLO – LEVEL 2

If you are aiming to achieve **Level 2**, you will need to **answer all four questions under Level 2** + the MATLAB question. The maximum mark you can achieve is **64.**

SOLO – Level 2: Question 1 (13 marks)

(a) An analogue signal x(t) = cos(650πt) + 2sin (720πt) is sampled 500 times per second. How many frequency components will the discrete-time signal obtained after sampling contain? Write down the expression for this discrete-time signal.

[4 marks]

(b) A difference equation for a particular filter is given by

$$y[n] = 0.12\,x[n] - 0.1\,x[n-2] + 0.82\,x[n-3] - 0.1\,x[n-4] + 0.12\,x[n-6]$$

(i) Find the impulse response of the above filter

[3 marks]

(ii) Using minimum number of multiplications, draw an implementation (a structure) of the filter given in (i) above.

[3 marks]

(c) If a sine wave of peak-to-peak amplitude 2A volts is quantised into N levels, show that the Signal to Quantisation Noise Ratio (SQNR) is given by

$$\text{SQNR (dB)} = 1.76 + 20 \log_{10}(N-1)$$

[3 marks]

SOLO – Level 2: Question 2 (13 marks)

(a) A first order filter has a transfer function given by $H(z) = \frac{1}{4}\,\frac{(1+z^{-1})}{(1-0.5z^{-1})}$. Draw the following realisation of H(z).

(i) Cascade **[2 marks]**

(ii) Parallel **[3 marks]**

(iii) Canonic **[2 marks]**

(b) Three systems with impulse responses $h_1[n] = \delta[n] - \delta[n-1]$, $h_2[n] = h_1[n]$ and $h_3[n] = u[n]$ are connected in cascade. Using the time domain convolution equation, calculate the impulse response, $h_c[n]$, of the overall system?

[3 marks]

(c) Sketch $y[n] = 2u[-n-3] - u[n+1]$; where $u[n]$is a unit step function

[3 marks]

SOLO – Level 2: Question 3 (13 marks)

(a) A first-order digital filter is described by the system function $H(z) = \frac{1}{3}\frac{(1+0.5z^{-1})}{(1-0.5z^{-1})}$

Compute the 3-dB cut-off frequency (θ_c) of the filter H(z). **[5 Marks]**

(b) Determine the transfer function H(z) of a discrete - time system with the pole-zero map shown in Figure 1:

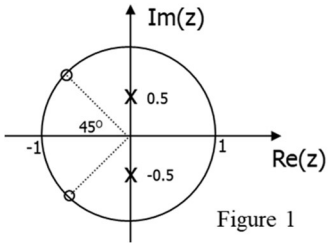


Figure 1

[4 Marks]

(c) Using the pole-zero map shown in part (b) above, sketch an approximate magnitude response of the H(z) **[4 Marks]**

SOLO – Level 2: Question 4 (13 marks)

(a) Determine and sketch the phase response of the following filter:

$$y[n] = x[n] + x[n-2]$$ **[5 Marks]**

(b) A 5 kHz sinusoidal is sampled at 40 kHz and 128 samples are collected and used to compute the 128 -point DFT of the signal.

(i) What is the time duration in seconds of the collected samples? **[2 Marks]**

(ii) At what DFT indices do we expect to see any peaks in the spectrum? **[3 Marks]**

(c) Show that the system specified by the pole-zero plot in Figure 2 is an all-pass filter, where a =0.5.

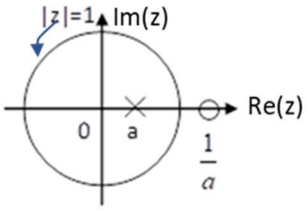


Figure 2

[3 Marks]

End of SOLO - Level 2 Examination Questions.

SOLO – LEVEL 3

If you are aiming to achieve **Level 3**, you will need to answer **any three Level 2 questions and one Level 3 question** + the MATLAB question. The maximum mark you can achieve is **74**.

SOLO – Level 3: Question (23 marks)

(a) A 90° phase shifter is a system H(z) with a frequency response

$$H(\theta) = \begin{cases} -j & 0 \le \theta \le \pi \\ j & -\pi \le \theta < 0 \end{cases}$$

(i) Show that the impulse response h[n] is given by

$$h[n] = \begin{cases} \dfrac{2}{n\pi} & n\ odd \\ 0 & n\ even \end{cases}$$

[8 marks]

(ii) Sketch the phase response of the system H(z)

[4 marks]

(b)

(i) Find the magnitude and phase of the system that has a unit sample (impulse) response:

$$h[n] = \delta[n] - \alpha\,\delta[n-1] \qquad \text{where } \alpha \text{ is real.}$$

[6 marks]

(ii) Using part (i) above, show that the system has a group delay (τ_g) given by the following equation:

$$\tau_g = \frac{\alpha^2 - \alpha\cos(\theta)}{1 + \alpha^2 - 2\alpha\cos(\theta)}$$

[5 marks]

End of SOLO - Level 3 Examination Question.

SOLO – LEVEL 4

If you are aiming to achieve **Level 4**, you will need to answer **any two Level 2 questions, one Level 3 question and one Level 4 question** + the MATLAB question. the maximum mark you can achieve is **84**.

SOLO – Level 4: Question 1 (23 marks)

(a) Determine the coefficients of a linear phase FIR filter

$$y[n] = b_0 x[n] + b_1 x[n-1] + b_2 x[n-2]$$

such that:

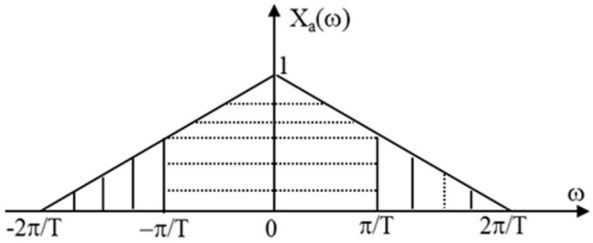
(i) It rejects completely a frequency component at $\theta_0 = \frac{2\pi}{3}$.

(ii) Its frequency response is normalised so that H(0) =1.

(b) Calculate and sketch the magnitude and phase response of the filter in part (a) above

[11 marks]

(c) For the system shown in Figure 3, $x_a(t)$ is the input signal. The analogue signal $x_a(t)$ has the spectrum $X_a(\omega)$ as shown below. $H(\theta)$ is a digital lowpass filter and is given by



$$H(\theta) = \begin{cases} 1 & |\theta| \le \frac{\pi}{4} \\ 0 & \frac{\pi}{4} < |\theta| \le \pi \end{cases}$$

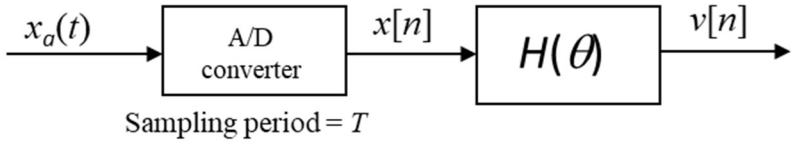


Figure 3

Sketch $|X(\theta)|$ and $|V(\theta)|$ against θ ; where θ is the digital frequency and ω is the analogue frequency.

[12 marks]

End of SOLO - Level 4 Examination Question.

SOLO – LEVEL 5

If you are aiming to achieve **Level 5**, you will need to **answer any one of the Level 2 questions, one Level 3 question, one Level 4 question and one Level 5 question +** the MATLAB question. The maximum mark you can achieve is **100**.

SOLO – Level 5: Question (29 marks)

(a) The Frequency response H(θ) of a filter is shown in Figure 5. Find the impulse response $h[n]$.

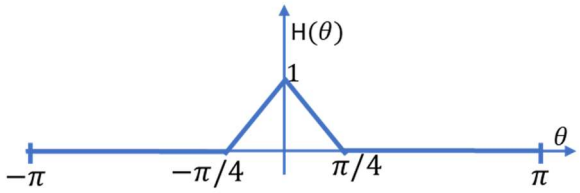


Figure 5

[10 marks]

(b) A stable and causal system (H(z)) may always be factored into a product of a minimum phase system with a linear phase system: $H(z) = H_{min}(z)$. $H_{lp}(z)$; where $H_{lp}(z)$ is a linear phase system.

$$H(z) = \frac{(2 - 3z^{-1})(3 + z^{-1})}{(1 - 0.5z^{-1})}$$

Find a minimum phase system function $H_{min}(z)$ and a linear phase system $H_{lp}(z)$, for the above H(z).

[8 marks]

(c) Consider an unstable digital filter whose transfer function is:

$$H(z) = \frac{(z+0.5)(2z+0.5)}{(z+5)(2z+5)}$$

A conceptual approach to stabilizing this filter is to cascade it with an all-pass filter $H_{ap}(z)$ (so as to preserve the magnitude response).

(i) What must be the transfer function $H_{ap}(z)$ of a filter such that the cascaded filter described by $H_s(z) = H(z).H_{ap}(z)$.
(ii) What is the transfer function $H_s(z)$ of the stabilised filter.
(iii) Is $H_s(z)$ causal? Minimum phase? Allpass?

[11 marks]

End of SOLO - Level 5 Examination Question.

MATLAB Question

You must answer the MATLAB question.
You should save your MATLAB program in the file *midexam_zID.m* (zID is your student zID) for uploading.

SOLO – Level 1: Question (12 marks)

A discrete time square wave signal 's' saved in a wav file '*sMeSquare.wav'* is located in folder 'Wav file' (Week 5 folder on Moodle). Write a MATLAB program to read the wav file to obtain the signal 's' and its sampling rate fs and then do the following tasks:

1. Filter the signal 's' using the filter $H(z)$ (Square to Sinusoid Converter) as shown in Figure 6 to obtain the sinusoid signal 'si' **[3 marks]**
2. Perform clipping operation on the signal 'si' to generate a signal 'sc', as per given equations in Figure 6. **[2 marks]**
3. Plot the first 600 samples of the signals 's', 'si' and 'sc'. The horizontal axis should be in milli second. **[2 marks]**
4. Compute and plot the magnitude spectrum of the signals 's', 'si' and 'sc' in the frequency region [0, fs/2]. The horizontal axis should be in Hertz. **[3 marks]**
5. Compute and plot the magnitude response of the filter H(z). The horizontal axis should be in Hertz. **[2 marks]**

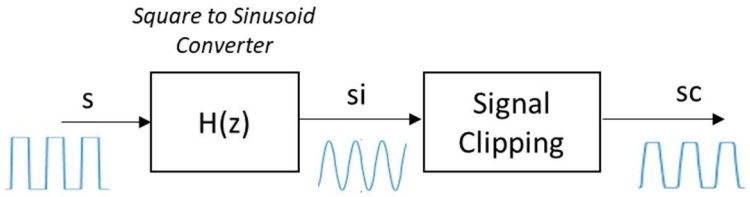


Figure 6

$$H(z) = \frac{0.0909 - 0.1605z^{-1} + 0.0736z^{-2}}{1 - 1.942z^{-1} + 0.9801z^{-2}}$$

$$\begin{cases} sc = si \ when \ |si| \leq 0.5 \\ sc = 0.5 \ when \ si > 0.5 \\ sc = -0.5 \ when \ si < -0.5 \end{cases}$$

Your program should plot all requested figures with the appropriate label on each axis including a title.

Hint: The in-built functions *audioread* could be used to read the wav file and its sampling rate